Enhanced Human Action Recognition from Synthetic UAVs Data Using Landmark Extraction and Mamba-based Model

Ninh Quang Tri, Luong Hoang Anh, Luu Quang Vu,
Tran Quoc Khoa, Phan Duy Hung

FPT University, Hanoi, Vietnam
trinqhe176305@fpt.edu.vn, anhlhhe176255@fpt.edu.vn,
vulqhe176902@fpt.edu.vn, khoatqhe170115@fpt.edu.vn,
hungpd2@fe.edu.vn

Abstract. Human action recognition from UAVs has become essential. Numerous studies have demonstrated that large datasets, including synthetic data, are essential for training human action recognition models. In this context, we introduce an approach that leverages synthetic data to enhance performance and competitiveness in the field, which can effectively address the data shortage problem from traditional methods. This work combines Mamba, an effective framework, with MediaPipe's pose estimation framework, significantly improving the extraction of skeletal key points from UAV footage. The process involves extracting crucial landmarks, calculating joint angles, and transforming these into a multivariate time series, which is then analyzed by a Mamba-based recognition model. The proposed method is tested on the RoCoG-v2 dataset with 57.5% accuracy - 17.3% higher than the state-of-the-art model. This result represents a significant advancement in human action recognition using UAVs, demonstrating the effectiveness of synthetic datasets in both research and real-world applications.

Keywords: Human action recognition, UAVs data, Mamba, RoCoG-v2, Mediapipe.

1 Introduction

Unmanned Aerial Vehicles (UAVs), commonly known as drones, have made great strides in recent years in many areas, such as filming [1], surveillance [2,3], search and rescue [4], and behavioral analysis [5]. Recently, one of the emerging applications of UAVs is in skeletal recognition [6]. This technology uses machine learning and computer vision methods to identify human postures and movements. UAVs, equipped with advanced imaging systems and powerful onboard processing capabilities, can capture high-resolution images and videos from various angles and altitudes [3]. This flexibility allows UAVs to monitor and analyze human activities in real-time, even in challenging environments where ground-based systems may be limited. The integration

of UAVs with human skeleton recognition algorithms enables the detection and tracking of human figures, the identification of specific postures, and the understanding of human behavior from an aerial perspective.

Recent research has focused on enhancing the accuracy and efficiency of these models. For instance, 2D pose estimation methods, which identify human joint coordinates in 2D images, are often used as a precursor to 3D pose estimation [7]. By converting 2D poses to 3D heatmap volumes, researchers can achieve more accurate and detailed skeletal models, which are crucial for applications requiring precise movement analysis [1,2].

This work presents an advanced human action recognition pipeline leveraging synthetic UAV data, integrating landmark extraction and a novel Mamba-based model (Fig. 1).

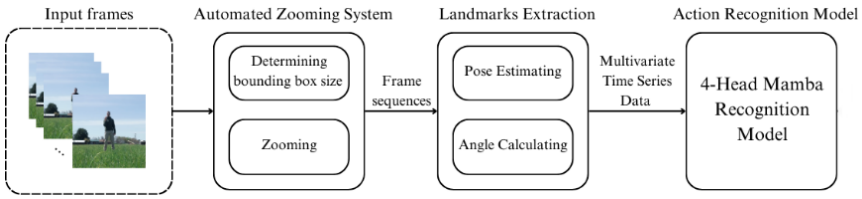

Fig. 1. Our pipeline uses synthetic UAV data with landmark extraction and a novel Mamba-based model. Innovations include a YOLOv5-powered Auto Zoom Algorithm for real-time subject tracking, Mediapipe-based joint angle extraction, and Mamba architecture for long-term dependency handling.

Our model has been rigorously tested on the RoCoG-v2 dataset, achieving a validation accuracy of 57.5%, which is 17.3% higher than the current state-of-the-art model, showcasing its superiority in action recognition tasks involving UAV-captured data.

2 Related Works

Human action recognition has evolved from early feature-based methods, such as Hidden Markov Models (HMMs) and Support Vector Machines (SVMs), which relied on hand-crafted features [9-11], to pose-based methods [12]. The revolution in 2010 introduced deep learning models, especially Convolutional Neural Networks (CNNs), which learned complex patterns from raw data [13]. Researchers also started incorporating multimodal fusion to improve performance [14]. Pose recognition identifies key joints of the human body (e.g., head, shoulders) from images or videos to create skeletal representations. Li et al. (2021) introduced a 2D pose estimation approach that detects key points in space [5]. OpenPose, a popular model, connects these key points to create a skeleton, performing well even in multi-person scenarios. Pavllo et al. (2019) proposed a 3D pose estimation model that uses temporal convolutions for video sequences, predicting 3D joint coordinates with semi-supervised

learning for improved accuracy [15]. The architecture is efficient, enabling real-time applications.

Liu and Szirányi (2021) presented a real-time human detection and gesture recognition system for UAV rescue operations. The system combines a modified YOLO for human detection and CNN-RNN layers for gesture recognition, achieving robust performance in complex environments [16]. This system enhances UAV-assisted rescue missions by quickly detecting and interpreting gestures. HRNet, introduced by Ke Sun et al. (2019), maintains high-resolution representations throughout the network, allowing accurate localization of body key points. It fuses multi-scale information, improving both local and global features, leading to state-of-the-art results on human pose estimation datasets [17]. Another method, using skeleton data represented as a Directed Acyclic Graph (DAG), predicts actions through a customized Directed Graph Neural Network (DGNN). Motion between frames is modeled for temporal dynamics, improving recognition accuracy by combining spatial and motion information [18].

2.1 RoCoG-v2 Dataset

In order to facilitate research in UAV-based action recognition and related fields, numerous public datasets have been developed for tasks such as human detection, object tracking, human re-identification, and action recognition [5, 19, 20]. Among these is RoCoG-v2 [21], a dataset created by Arun V. Reddy et al. which comprises 107,458 videos, including 106,996 synthetic videos and 482 real-world videos, focusing on seven action classes derived from the U.S. Army Field Manual. These action classes include gestures such as "Follow Me," "Advance," "Halt," "Rally," "Attention," "Move Forward," and "Move in Reverse," as illustrated in Figure X. The dataset encompasses both ground and aerial perspectives, with real videos collected from outdoor environments and synthetic data generated using Unity. Despite its relevance for studying the integration of real and synthetic data, prior research has found it challenging to achieve high accuracy when validating models trained on synthetic data with real-world test sets, highlighting the ongoing limitations in effectively generalizing synthetic data for real-world applications. RoCoG-v2 remains a critical resource for applications in action recognition, auto-zooming, and temporal reasoning, but it underscores the need for improved approaches to bridge the gap between synthetic and real data.

2.2 YOLOv5, Mediapipe and Mamba

In recent years, advancements in object detection and human action recognition have seen significant contributions from various models. YOLOv5 [22] is among the leading object detection frameworks, known for its high speed and accuracy in real-time applications. It employs a deep convolutional neural network to detect objects within images or video frames, producing precise bounding boxes across multiple classes. This efficiency is particularly beneficial for UAV systems, where quick identification and tracking of moving targets are essential [23,24]. YOLOv5 has been effectively used in

numerous applications, including gesture recognition [25] and behavior analysis [26], providing a robust foundation for detecting human actions in aerial surveillance contexts.

In parallel, skeleton-based action recognition methods have gained prominence due to their effectiveness in capturing human motion dynamics. Mediapipe [27], developed by Google, provides real-time pose estimation capabilities by detecting key body joints and constructing a skeletal diagram. This framework enables the tracking of joint angles and body posture changes, which are critical for extracting meaningful motion cues for downstream tasks such as temporal reasoning and action classification [28,29]. Mediapipe's versatility makes it suitable for applications involving UAV monitoring and gesture-based control systems, enhancing the robustness of action recognition pipelines.

Furthermore, the Mamba model [30] offers a novel approach to modeling time-series data, designed to efficiently capture long-term dependencies in action sequences. By utilizing a structured State Space Model (SSM) [31] combined with selective attention mechanisms, the Mamba model addresses the challenges of traditional architectures, such as Transformers, while maintaining computational efficiency [32,33]. This architecture is particularly well-suited for dynamic scenarios involving complex human behaviors, improving the understanding of actions over extended periods.

3 Methodology

We present a pipeline consisting of multiple stages from automatic zooming, landmark extraction, angle calculation, training and testing of the recognition model. The input for our method is the sampled video frames. This pipeline will first go through the phase of identifying and zooming in on the object that needs to be action recognized. After that, we employ MediaPipe [27] for landmarks extraction from the frames, calculating angles between these landmarks to form a multivariate time series data. These data will be used for action recognition by utilizing a sophisticated Mamba-based recognition model.

3.1 Automated Zooming System

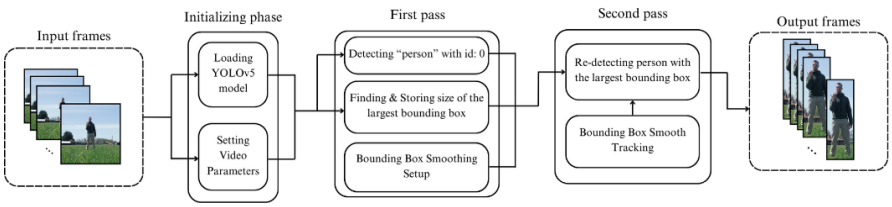

Fig. 2. Automated Zooming System using YOLOv5.

The auto-zooming mechanism developed here adjusts the view based on objects detected by the pre-trained YOLOv5 [22] model in real-time video. As shown in Fig 2,

the process starts with detecting an object of interest, specified by the user. Once YOLOv5 identifies and localizes the object with a bounding box, the system uses these coordinates to adjust zoom. A key feature, Region of Interest (RoI) Centralization, ensures the detected object is positioned centrally for clear visibility. When objects are near the frame edges, the system re-centers them to maintain prominence in the visual field. Adaptive zooming adjusts zoom levels based on the object's size and position—zooming in for smaller, off-center objects and minimal adjustments for larger, well-centered ones—ensuring consistent visibility without overwhelming the frame. Smooth zoom transitions prevent abrupt changes, using incremental adjustments to maintain clarity in real-time. For optimized performance, frame skipping reduces processing demands, and GPU acceleration speeds up detection and zooming, supporting responsiveness without sacrificing accuracy.

3.2 Landmark Extraction

Mediapipe is used to extract human poses within detected bounding boxes for detecting body landmarks necessary for pose analysis. Mediapipe's Pose model was applied to each cut frame of the recorded video sequences. The model leverages a deep learning framework to identify 33 key landmarks on the human body as in Fig 3. Mediapipe's Pose model [27] processes these even under varied conditions such as changing lighting or complex backgrounds, while maintaining reliable accuracy. Each identified landmark provided precise coordinates that map the joint in three-dimensional space for relevant data from the subject's movements.

Fig. 3. Skeleton-based human keypoints.

Following landmark extraction, 14 specific angles computed in radian required for this study's analysis will be computed. These angles were determined by selecting sets of three landmarks forming each joint or limb segment to calculate angular measurements that reflect joint orientations. These angles will replace the coordinates of landmarks which are susceptible to jittering caused by camera or human's movement for the recognition model's input.

These calculated angles were organized into sequences to represent each frame in a time-series format. Interpolation was applied to standardize sequence lengths across frames. Additionally, noise reduction and filtering were conducted to smooth the data and mitigate inconsistencies due to slight inaccuracies in landmark detection.

3.3 Mamba-based Action Recognition Model

We propose the Multi-Head Mamba Classification Model - a specialized architecture designed to handle multivariate time series data extracted from the previous step by leveraging multiple processing heads of Mamba model architecture [30]. As shown in Fig 4., experiments were carried out using 4 heads. This multi-head structure enables the model to capture different aspects or representations of the input data simultaneously, enhancing its ability to extract meaningful features. Additionally, a Cosine Similarity Classifier module is applied for determining the similarity score between the combined feature vector and learned class prototypes allowing a threshold to be defined for an extra class of 'Unrecognized' in a real-time environment.

All components of the classification model will be trained simultaneously with the extracted angle sequences. The loss function, cross-entropy for multi-class classification tasks, is minimized using the Adam optimization algorithm [34].

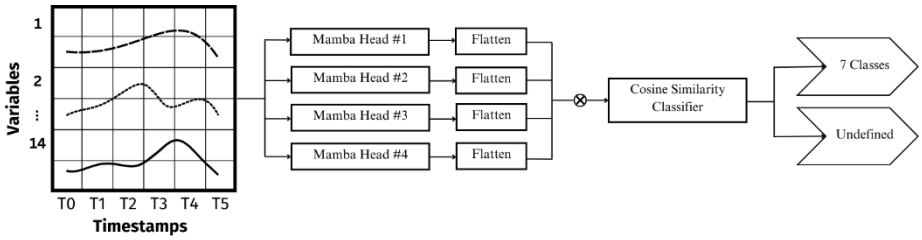

Fig. 4. The 4-Head Mamba Classification Model classifies input from the previous step into 7 base classes of the RoCoG-v2, with the addition of the class 'Unrecognized'.

Mamba Head

The Mamba architecture is a linear-time sequence model designed to address some limitations of traditional models like Transformers [8]. Excelling in modeling long-range dependencies within sequences, a capability attributed to its parametric design that facilitates the retention and utilization of information across extended sequences. This feature is particularly beneficial for tasks requiring the analysis of complex temporal relationships and causal sequence modeling. Mamba is built on selective State Space Models (SSMs), which combine the strengths of RNNs and CNNs but add an input-dependent selection mechanism to enhance content-based reasoning [30].

Mamba's robust handling of long sequence information makes it suitable for maintaining temporal coherence and modeling extended dependencies. Therefore, the usage of Mamba in the recognition model architecture due to its ability to effectively capture the temporal information in sequence of motion, enhancing accuracy in action recognition problems.

The outputs of each Mamba Head are flattened and concatenated into a feature vector. This step combines the learned features of the independent Mamba Heads into one unified representation.

Cosine Similarity Classification

The Cosine Similarity Classifier module is designed to classify input feature vectors based on their cosine similarity to class prototypes. The main idea behind using cosine similarity lies in its ability to measure the directional similarity between vectors, regardless of their magnitude, making it robust to variations in feature scales. In the module's forward pass, input vectors are normalized to unit vectors using L2 normalization to ensure the classification is based purely on the direction of each vector in a high-dimensional space. By computing the cosine similarity between the normalized input vector and the normalized class prototypes, which are represented by the transpose of the weight matrix, the module returns a set of similarity scores, each corresponding to a class. The cosine similarity between the input vector and each normalized weight vector is computed as:

$$\text{cosine_similarity}_{i,j} = \frac{x_i \cdot W_j^*}{\|x_i\|_2 \left\|W_j^*\right\|_2} = \frac{\sum_{k=1}^{d} x_{i,k} W_{j,k}^*}{\sqrt{\sum_{k=1}^{d} {x_{i,k}}^2} \sqrt{\sum_{k=1}^{d} {W_{i,k}^*}^2}}$$

Let x_i be the i-th input vector with dimension d, W^* be the weight matrix with shape $d \times C$, where C is the number of classes, the j-th column of the weight matrix, representing the weight vector for class j. Each input vector x_i and each column W^* of the weight matrix is normalized to a L2 norm unit.

4 Results

While synthetic data has emerged as a promising approach for training models efficiently, its application in real-world scenarios remains limited, as significant discrepancies often arise between synthetic and real-world datasets. Prior research in human action recognition has repeatedly struggled to bridge this gap, with accuracy typically falling short when models trained on synthetic data are validated against real-world observations [35,36]. To further investigate these limitations, our approach will be tested by training exclusively on synthetic data while validating on real-world data. This setup aims to evaluate the model's ability to generalize across the synthetic-real domain gap, offering insights into how effectively knowledge gained from synthetic environments can transfer to real-world applications.

The accuracy of the model over the training steps is depicted in Fig 5. The training accuracy shows a steady increase, reaching a peak value of approximately 0.81. In contrast, the testing accuracy exhibits fluctuations and reached its peak at 0.5759, shows a large gap between training and testing accuracies, indicating overfitting. This likely stems from using synthetic data, which lacks the complexity of real-world data.

Consequently, the model achieves high accuracy on synthetic data but struggles with more variable real data, as seen in moderate testing accuracy.

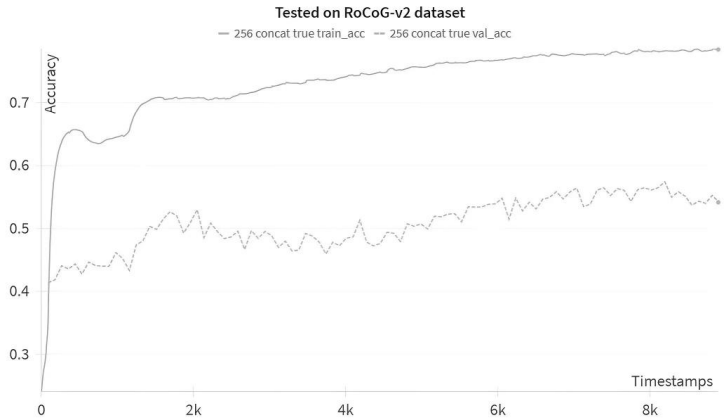

Fig. 5. Accuracy curve over step (best 0.5759 on test data).

Table 1 provides a comparison between different methods on the RoCoG-v2 dataset, focusing on test accuracy, the algorithms used, and the optimization techniques. The method introduced by Dan et al. [35] employs the MoViNet A0 and A2 algorithms, both optimized with RMSprop, yielding test accuracies of 32.8% and 34.1%. The method proposed by Xiyun et al. [36] employs the AZTR algorithm with Stochastic Gradient Descent (SGD) as the optimizer, achieving a test accuracy of 40.2%. In comparison, methods based on Transformer and Long Short-Term Memory (LSTM) [36] architectures, both using Adam optimizers, show incremental improvements with accuracies of 43.13% and 48.75%, respectively. The highest performance comes from our Mamba-based model, which also uses the Adam optimizer. This approach significantly surpasses the others, reaching a test accuracy of 57.59%. This indicates that our Mamba algorithm utilizing cosine similarity, combined with the Adam optimizer, offers substantial improvements in human action recognition from UAV data, particularly when compared to other state-of-the-art methods.

Table 1. Comparison of our method and others on the RoCoG-v2 dataset

Method	Algorithms	Optimizer	Accuracy
Dan et.al, 2021 [35]	MoViNet A0	RMSprop	32.8%
Dan et.al, 2021 [35]	MoViNet A2	RMSprop	34.1%
Xiyun et al., 2023 [36]	AZTR	SGD	40.2%
Our method	Transformer [8]	Adam	43.13%
Our method	LSTM [37]	Adam	48.75%
Our method	**Mamba**	**Adam**	**57.59%**

5 Conclusion

This research introduces a novel approach to human action recognition from UAV data by integrating landmark extraction with a Mamba-based model. Key contributions include: (1) an automated zooming system using YOLOv5 for dynamic human tracking, improving data quality for action recognition, (2) the use of MediaPipe for effective landmark extraction and angle-based skeleton modeling, making recognition robust to lighting and perspective changes, and (3) a new Mamba-based model with cosine similarity classification, which achieved a 57.5% validation accuracy on the RoCoG-v2 dataset, outperforming prior methods by 17.3%. Despite these advancements, challenges remain, such as overfitting and inconsistent class performance, highlighting opportunities for future research in domain adaptation and hybrid architectures to enhance UAV-based human action recognition further.

References

1. Pitas, I., Mademlis, I.: Autonomous UAV Cinematography. In: Proceedings of the 30th ACM International Conference on Multimedia (MM '22), pp. 7374–7376. Association for Computing Machinery, New York, NY, USA (2022)
2. Fang, Z., Savkin, A.V.: Strategies for Optimized UAV Surveillance in Various Tasks and Scenarios: A Review. Drones **8**, 193 (2024)
3. Sun, Y., Zhi, X., Han, H., Jiang, S., Shi, T., Gong, J., Zhang, W.: Enhancing UAV Detection in Surveillance Camera Videos through Spatiotemporal Information and Optical Flow. Sensors (Basel) **23**(13), 6037 (2023)
4. Papić, V., Gudelj, A.Š., Milan, A., Miličević, M.: Person Mobility Algorithm and Geographic Information System for Search and Rescue Missions Planning. Remote Sensing **16**(4), 670 (2024)
5. Li, T., Liu, J., Zhang, W., Ni, Y., Wang, W., & Li, Z.: UAV-Human: A large benchmark for human behavior understanding with unmanned aerial vehicles. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Nashville, TN, USA, pp. 16261–16270. IEEE Computer Society, Los Alamitos, CA, USA (2021)
6. Pham, D.-T., Hoang, V.-N., Le, V.-D., Nguyen, T.-T., Tran, T.-H., Vu, H., Le, V.-H., Le, T.-L.: Deep Learning Models for Skeleton-Based Action Recognition for UAVs. In: Proceedings of the IEEE Ninth International Conference on Communications and Electronics (ICCE), pp. 343–348. Nha Trang, Vietnam (2022)
7. Chen, H., Feng, R., Wu, S., Xu, H., Zhou, F., Liu, Z.: 2D Human Pose Estimation: A Survey. Multimedia Systems **29**(5), 3115–3138 (2023)
8. Vaswani, A. et al. : Attention is All You Need. In: Advances in Neural Information Processing Systems, vol. 30 (2017)
9. Wahid, M.F., Shahriar, M.F., Sobuj, M.S.I.: A Classical Approach to Handcrafted Feature Extraction Techniques for Bangla Handwritten Digit Recognition. In: Proceedings of the International Conference on Electronics, Communications and Information Technology (ICECIT), pp. 1–4. Khulna, Bangladesh (2021)
10. Afsar, P., Cortez, P., Santos, H.: Automatic Human Action Recognition from Video Using Hidden Markov Model. In: Proceedings of the IEEE 18th International Conference on Computational Science and Engineering, pp. 105–109. Porto, Portugal (2015)

11. Chathuramali, K.G., Rodrigo, R.: Faster Human Activity Recognition with SVM. In: Proceedings of the International Conference on Advances in ICT for Emerging Regions (ICTer2012), pp. 197–203. Colombo, Sri Lanka (2012)
12. Wang, C., Wang, Y., Yuille, A.L.: An Approach to Pose-Based Action Recognition. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 915–922. Portland, OR, USA (2013)
13. Ali, A., Pinyoanuntapong, E., Wang, P., Dorodchi, M.: Skeleton-based Human Action Recognition via Convolutional Neural Networks (CNN). arXiv:2301.13360 (2023)
14. Ahmad, M., Lee, S.-W.: HMM-based Human Action Recognition Using Multiview Image Sequences. In: Proceedings of the 18th International Conference on Pattern Recognition (ICPR'06), pp. 263–266. Hong Kong, China (2006)
15. Pavllo, D., Feichtenhofer, C., Grangier, D., Auli, M.: 3D Human Pose Estimation in Video with Temporal Convolutions and Semi-Supervised Training. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 7745–7754. Long Beach, CA, USA (2019)
16. Liu, C., Szirányi, T.: Real-Time Human Detection and Gesture Recognition for On-Board UAV Rescue. Sensors **21**(6), 2180 (2021)
17. Sun, K., Xiao, B., Liu, D., Wang, J.: Deep High-Resolution Representation Learning for Human Pose Estimation. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 5686–5696. Long Beach, CA, USA (2019)
18. Shi, L., Zhang, Y., Cheng, J., Lu, H.: Skeleton-Based Action Recognition with Directed Graph Neural Networks. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 7904–7913. Long Beach, CA, USA (2019)
19. Lyu, Y., Vosselman, G., Xia, G.S., Yilmaz, A., Yang, M.Y.: UAVid: A Semantic Segmentation Dataset for UAV Imagery. ISPRS Journal of Photogrammetry and Remote Sensing **165**, 108–119 (2020)
20. Perera, A.G., Law, Y.W., Chahl, J. (2019). UAV-GESTURE: A Dataset for UAV Control and Gesture Recognition. In: Leal-Taixé, L., Roth, S. (eds) Computer Vision – ECCV 2018 Workshops. ECCV 2018. Lecture Notes in Computer Science(), vol 11130. Springer, Cham
21. Reddy, A.V., et al.: Synthetic-to-Real Domain Adaptation for Action Recognition: A Dataset and Baseline Performances. In: Proceedings of the IEEE International Conference on Robotics and Automation (ICRA), pp. 11374–11381. London, United Kingdom (2023)
22. Jocher, G., et al.: Ultralytics/yolov5: V7.0 - YOLOv5 SOTA Realtime Instance Segmentation. Zenodo (2022)
23. Yang, X., Xiu, J., Liu, X.: Research on Improved YOLOv5 Vehicle Target Detection Algorithm in Aerial Images. Drones **8**(5), 202 (2024)
24. Gao, S., Gao, M., Wei, Z.: MCF-YOLOv5: A Small Target Detection Algorithm Based on Multi-Scale Feature Fusion Improved YOLOv5. Information **15**(5), 285 (2024)
25. Reddy, V.P., Reddy, V.V., Sukriti: Sign Language Recognition Based on YOLOv5 Algorithm for the Telugu Sign Language. arXiv:2406.10231 (2024)
26. Yang, F., Wang, X.: SCB-ST-Dataset4: Extending the Spatio-Temporal Behavior Dataset in Student Classroom Scenarios Through Image Dataset Method. arXiv:2310.16267 (2023)
27. Lugaresi, C., Tang, J., Nash, H., McClanahan, C., Uboweja, E., Hays, M., Grundmann, M.: Mediapipe: A framework for building perception pipelines. arXiv:1906.08172 (2019)
28. Abedi, A., Khan, S. S.: Engagement Measurement Based on Facial Landmarks and Spatial-Temporal Graph Convolutional Networks. arXiv:2403.17175 (2024)
29. Putra, I. A., Nurhayati, O. D., Eridani, D.: Human Action Recognition (HAR) Classification Using MediaPipe and Long Short-Term Memory (LSTM). TEKNIK **43**(2), 190–201 (2022)

30. Gu, A., & Dao, T.: Mamba: Linear-time sequence modeling with selective state spaces. arXiv:2312.00752 (2023)
31. Gu, A., Goel, K., & Ré, C.: Efficiently modeling long sequences with structured state spaces. arXiv:2111.00396 (2021)
32. Zhu, C., Gao, S., Chen, H., Guo, G., Wang, C., Wang, Y., & Fan, Q.: Why mamba is effective? Exploit Linear Transformer-Mamba Network for Multi-Modality Image Fusion. arXiv:2409.03223 (2024)
33. Qu, H., Ning, L., An, R., Fan, W., Derr, T., Xu, X., & Li, Q.: A survey of mamba. arXiv:2408.01129 (2024)
34. Kingma, D., & Ba, J.: Adam: A method for stochastic optimization. arXiv:1412.6980 (2014)
35. Kondratyuk, D., Makarov, I., Blokhin, A., Trunov, D., & Mukhin, A.: MoViNets: Mobile Video Networks for Efficient Video Recognition. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 16015–16025. Nashville, TN, USA (2021)
36. Wang, X., Yao, Y., Huang, Y., Yang, Y., & Liu, X.: AZTR: Aerial Video Action Recognition with Auto Zoom and Temporal Reasoning. In: Proceedings of the IEEE International Conference on Robotics and Automation (ICRA), pp. 1312–1318 (2023)
37. Hochreiter, S., & Schmidhuber, J.: Long Short-Term Memory. Neural Computation 9(8), 1735–1780 (1997)